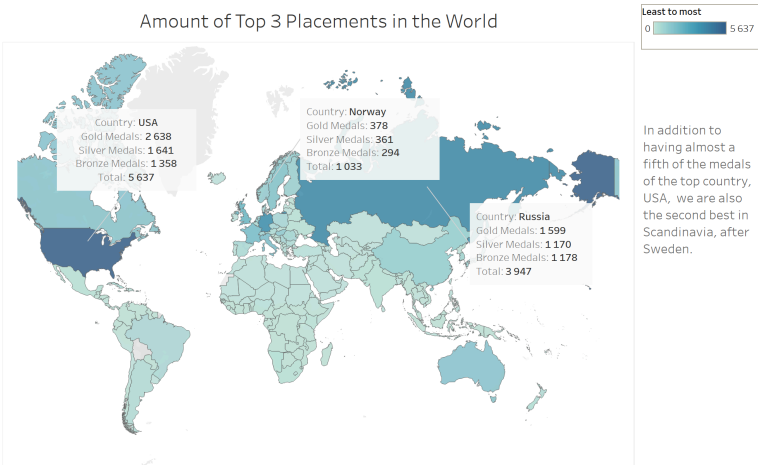
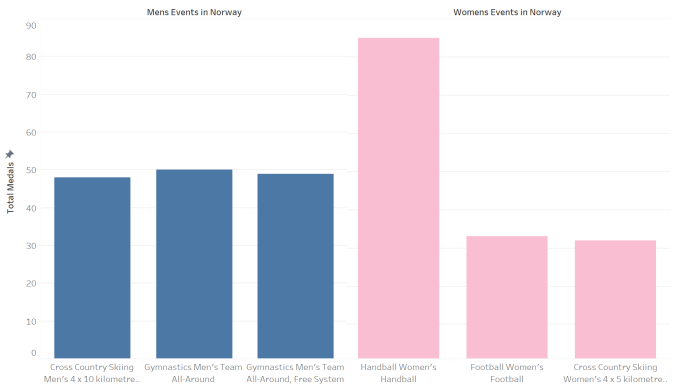


Static Explanatory Data Visualisation

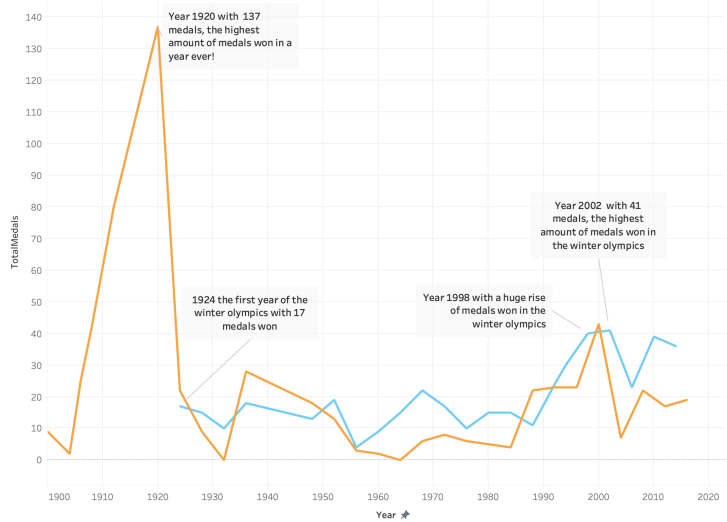
Amount of Top 3 Placements in the World



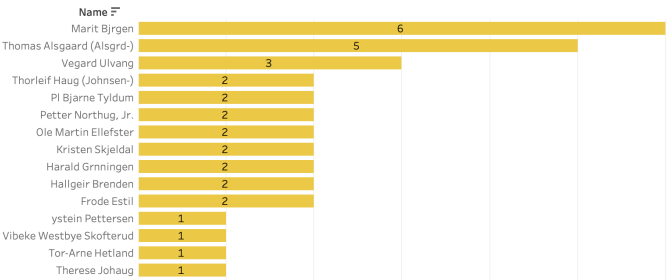
Top 3 Events by Amount of Medals Won for Norwegian Athletes



Norway medals summer vs winter



Top 15 norwegian athletes with the most gold medals in Cross Country Skiing



Explanatory Text

Data Preparation

After some brainstorming on how we wanted to go about this, we started off by refactoring the first data-set (Athletes-table) from CSV-format to XLSX-format, using Excel. To be able to create a relationship between the two tables we also had to import the other data-set (Regions-table) the same way. When we began to extract some of the data we quickly realised we had to make several calculated fields and parameters to narrow down which of the nominal and discrete data we wanted to work with, and removing any unnecessary data. This was a bit challenging at first but the more we got to know the data, the easier it was to debug anything that looked wrong, like medal count or when comparing the athletes by gender. The names of the Norwegian athletes were also spelled wrong which made our search function sort of difficult, since the user is probably gonna type it correctly.

Relevance to the Case

Since our targeted audience were the Norwegian athletes and coaches we wanted to focus mainly on the athlete's historical performance and medals, to motivate them before heading in to practice. We decided that in the interactive dashboard it would be efficient to create a mix of explanatory and exploratory charts, so that the user would be able to play around with some of the data, but also motivating them with annotations that highlighted their great performance.

We first created a map to get an overview of all the region's performance in the Olympic Games based on their medals, and when hovering over each country the user is able to see how many medals they each have. They can also filter the entire map by medal type, which changes the gradient by which country has the most medal in the type of medal that is selected. In the static version we chose a few countries to compare Norway with, and displayed their amount of medals.

We also wanted a chart displaying the gender difference in the different sports that Norway competes in, so the user is able to choose which sport to display. For the static version we chose to show the top 3 events, where each gender has the most medals.

We also chose a time series graph to display the difference in medals in both summer- and winter-olympics. This helped us create a picture of when the winter-olympics started, as it didn't start until the very beginning of 1924 compared to summer-olympics that started as early as in 1896. The time series graph clearly shows how Norway has improved over time. and the user is able to use the dashboard to compare every region's performance (by medals), sorting by season and which sport. As for the static dashboard we again only wanted to focus on Norway, and therefore used annotations to highlight the years where huge milestones were accomplished. This tells the user that although Norwegians previously performed best at the summer-olympics, the stats for winter-olympics are continually rising and has even surpassed when looking at the medals they've won so far.

Explanation of Design Choices

We wanted the user to be able to find a specific athlete by name, sort by region, filter through the different sports and choose which medal type to display. To make this possible we made a perimeter that excluded the sports where none of the Norwegian athletes had won a medal. We sorted all the athletes into "Norwegian" and "Others" based on the NOC being 'NOR' - so that we could display only the Norwegians in some charts, and still have the option to display all when needed to compare.

For the map we used a gradient with a blue hue, where the darker shades represent more medals, and the lighter shades represent fewer medals - in both dashboards. In the static version we also added explanatory annotations for

both the US and Russia as well as Norway, to display the differences in performance in Norwegian athletes compared to bigger countries. We chose the caption based on what we want the users to focus on.

For the gender comparison we used the stereotypical colors that most people associate with the two biological genders, male and female, however we don't think certain colors have anything to do with a person's gender. This was only used because a lot of text can be hard to read when it's that small, and when the user is just passing by quickly (Kirk, p.256).

Although historical data doesn't change over time, and are rather a one-time statement per olympic game, we still decided that a line chart would create a good visual on the data we wanted to represent. It's an effective way to see how the athletes have improved and progressed over time. Cole Knafllic states that line charts are most commonly used for continuous data. This is often in some measure of time such as years (Knafllic, p.51) as we used. To display the precise amount of medals won each year we first used labels, but quickly saw how cluttered it was with all the numbers. «*Visual clutter creates excessive cognitive load and can hinder the message*» (Knafllic, p.74). Therefore we chose to leave the chart as it was as the y-axis already tells us the same information.

It was easy to choose the bar chart to show a clear visual on how many gold, silver and bronze medals each Norwegian athlete has won, since this is a categorical chart. It's easy to read for almost anyone, even when containing a lot of values, and since there were multiple categories (medals, athletes, names) to display, the bar chart had an overall best outcome for this. There was never any doubt on what colors that were gonna represent each medal, since most people associate the different medal types to the color of the metal (alloy) itself (Kirk, p. 256). However this turned out to be a little difficult when we wanted to sort them by their "value". Tableau ended up prioritizing the amount of medals over the value of the medals itself, which put people with bronze medals over silver medals etc. We explored different ways to solve this, like giving each medal a significant value and then sorting them by said value, but nothing we found worked. We decided to keep it as is, since it only showed a false visual in the sports that had fewer participants.

When choosing the titles, headers and basically most of our text, we took inspiration from Andy Kirk's four categories: statement title, question title, descriptive title and artistic titles (Kirk, p. 231). Most of our titles are short and concise to help us convey the motivating parts more efficiently to the user, and therefore the static dashboard has a bit more text for even more explanation. In other words we mainly used statements and descriptive titles/texts in order to get the user intrigued and wanting to read more.

As for the composition of the dashboard itself we knew it would be displayed on a bigger screen, and therefore chose a design to convey motivating insights into the athlete's performance, without overwhelming the user. A lot of the data is self explanatory just by how the visuals are set up, which means that other people with little to none prior knowledge of the Olympic Games, could still look at this and know exactly (or at least have a clue of) what they're looking at. We were happy that we didn't have to compromise as much when it came to displaying the different charts on the dashboard, as we would have to do if it was being displayed on a smaller screen. We did however need to make some compromises when making the static version, such as only showing the most important data and using multiple annotations to highlight the athlete's performance in the amount of medals they have, for the user in a explanatory but still static visual (Kirk, p. 288).

Bibliography

- Kirk, Andy, *Data Visualisation: A Handbook for Data Driven Design*, 2nd edition, Sage Publications 2019
- Knafllic, Cole, *A Data Visualization Guide for Business Professionals*, John Wiley & Sons 2015

Member Contributions

All members contributed equally to the exam. Everyone made one visual each for both the interactive and static dashboards, and we collaborated on the last charts. Everyone also helped each other a lot, and asked for help when needed. Each candidate wrote about their own charts in the report before we merged it all together for one final version. Everyone in the group was willing to meet physically multiple times per week, and we have also been communicating online from home.